THE COLLISION SPECTRUM

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Abstract. For a prime base b, working at lag $\ell = 1$ (modulus $m = b^2$), and a primitive odd Dirichlet character χ modulo b^2, the collision transform coefficient $\hat{S}^\circ(\chi)$ admits an exact factorization:

$$\hat{S}^\circ(\chi) = -\frac{B_{1,\overline{\chi}} \cdot \overline{S_G(\chi)}}{\phi(b^2)},$$

where $B_{1,\overline{\chi}}$ is the generalized first Bernoulli number and $S_G(\chi)$ is the diagonal character sum. By the standard Bernoulli–L-value formula, $|B_1| = (b/\pi)\,|L(1,\chi)|$, so the collision invariant's Fourier spectrum encodes L-function special values.

A Parseval identity gives an exact formula for the weighted second moment $\sum |L(1,\chi)|^2 \cdot |S_G(\chi)|^2$ in terms of the collision invariant's values on the finite group. The digit function computes this L-value moment exactly. Under a conditional zero-free hypothesis, the triangle inequality yields a separate bound connecting $L(1)$ to $L(s)$ for s in the critical strip.

At base 5, the factorization gives $|\hat{S}^\circ| \propto |L(1)|^2$ exactly. For quadratic characters in the family, the decomposition specializes to class-number data.

1. Introduction

The collision invariant S_ℓ, introduced in [4], is a function on the finite group $(\mathbb{Z}/m\mathbb{Z})^\times$ with $m = b^{\ell+1}$. The collision transform of [5] decomposes S_ℓ over Dirichlet characters. The antisymmetry theorem restricts the centered decomposition to odd characters, and the convergence theorem proves the resulting prime harmonic sum $F^\circ(1)$ is finite.

This paper identifies what the collision transform encodes. The collision spectrum is not merely correlated with L-function values. It is built from them. The main result is the decomposition theorem: each Fourier coefficient factors into a generalized Bernoulli number (encoding $|L(1)|$) and a diagonal character sum. The proof uses the slice formula from [4], the classical Bernoulli identity for character sums over fractional parts, and the vanishing of coset sums for primitive characters.

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2. The Diagonal Character Sum

Lemma 1 (Diagonal reduction)**.** *For any base $b \geq 2$ and any odd Dirichlet character χ modulo b^2:*

$$S_G(\chi) = -2\,\overline{\chi}(b{+}1)\sum_{k=1}^{b-1}\overline{\chi}(k).$$

In particular, $|S_G(\chi)| = 2\left|\sum_{k=1}^{b-1}\overline{\chi}(k)\right|$.

Proof. Write $S_G = Q - \alpha P$ where $Q = \sum_{r=0}^{b-2}\overline{\chi}(r(b{+}1){+}1)$ (the $n{+}1$ terms, using $\overline{\chi}(b^2) = 0$), $\alpha = \overline{\chi}(b{+}1)$, and $P = \sum_{k=1}^{b-1}\overline{\chi}(k)$ (the n terms, using multiplicativity $\overline{\chi}(r(b{+}1)) = \alpha\,\overline{\chi}(r)$ and $\overline{\chi}(0) = 0$).

The reflection $b^2 - (r(b{+}1){+}1) = (b{-}1{-}r)(b{+}1)$ maps $\{r(b{+}1){+}1 : r = 0, \ldots, b{-}2\}$ bijectively onto $\{j(b{+}1) : j = 1, \ldots, b{-}1\}$. Since χ is odd, $\overline{\chi}(b^2{-}a) = -\overline{\chi}(a)$ for every a. Applying this to each term of Q:

$$-Q = \sum_{j=1}^{b-1}\overline{\chi}(j(b{+}1)) = \alpha\sum_{j=1}^{b-1}\overline{\chi}(j) = \alpha P.$$

Therefore $Q = -\alpha P$ and $S_G = -\alpha P - \alpha P = -2\alpha P$. □

3. The Decomposition Theorem

Theorem 2 (Decomposition)**.** *Let b be prime, $m = b^2$, and χ a primitive odd character modulo m. Then*

$$\hat{S}^{\circ}(\chi) = -\frac{B_{1,\overline{\chi}}\cdot\overline{S_G(\chi)}}{\phi(m)},$$

where $B_{1,\overline{\chi}} = (1/m)\sum_a a\,\overline{\chi}(a)$ and $S_G(\chi) = \sum_{n\in G}[\overline{\chi}(n{+}1) - \overline{\chi}(n)]$, and $G = \{n \in \{0, \ldots, m{-}1\} : \lfloor n/b\rfloor = n \bmod b\}$ is the diagonal set [4] *(elements whose base-b digits coincide), with $|G| = b$. Dirichlet characters are extended by $\chi(a) = 0$ for $\gcd(a, m) > 1$.*

Proof. Write $\phi = \phi(m)$. Expand $\phi\,\hat{S}^{\circ} = \sum_a S^{\circ}(a)\,\overline{\chi}(a)$ using the slice formula [4]: $S(a) = -1 - \lfloor a/b\rfloor + \sum_{n\in G} d_n(a)$ where $d_n(a) = \lfloor (n{+}1)a/m\rfloor - \lfloor na/m\rfloor$.

Step 1 (centering vanishes). Throughout, $m = b^2$ (lag 1). The spectral classes are the cosets $\{a : a \equiv k \pmod b\}$ for coprime k [5]. The class mean $\overline{S}_R$ is constant on each coset.

For primitive χ modulo b^2: the restriction of χ to the subgroup $H = \{u \in (\mathbb{Z}/b^2\mathbb{Z})^* : u \equiv 1 \pmod b\}$ is non-trivial (otherwise χ would factor through $(\mathbb{Z}/b\mathbb{Z})^*$, contradicting primitivity). Therefore

$\sum_{u\in H}\overline{\chi}(u)=0$, and since the coset $\{a\equiv k \pmod b\}$ is a translate $k\cdot H$:

$$\sum_{\substack{a \bmod b^2\\ a\equiv k \bmod b}}\overline{\chi}(a)=\overline{\chi}(k)\sum_{u\in H}\overline{\chi}(u)=0.$$

The centering correction $\sum_a \overline{S}_{R(a)}\,\overline{\chi}(a)=\sum_R \overline{S}_R\sum_{a\equiv R+1 \bmod b}\overline{\chi}(a)=0$.

Step 2 (constant and floor terms). $\sum_a(-1)\,\overline{\chi}(a)=0$. The fractional part $\{a/b\}$ depends only on $a \bmod b$, so

$$\sum_a \overline{\chi}(a)\,\{a/b\}=\sum_{k=1}^{b-1}\frac{k}{b}\sum_{a\equiv k \bmod b}\overline{\chi}(a)=0,$$

since each coset sum vanishes (Step 1). Therefore $-\sum\lfloor a/b\rfloor\,\overline{\chi}(a)=-b\,B_{1,\overline{\chi}}$.

Step 3 (diagonal terms). The endpoints $n=0$ and $n=m-1$ contribute nothing: $d_0(a)=0$ (since $\lfloor a/m\rfloor=0$), and $d_{m-1}(a)=1$ for all units, giving $\sum_a\overline{\chi}(a)=0$. For the remaining $n=r(b+1)$ with $1\le r\le b-2$: both $n\equiv r\not\equiv 0 \pmod b$ and $n+1\equiv r+1\not\equiv 0 \pmod b$, so $\gcd(n,m)=\gcd(n+1,m)=1$ and Lemma 3 applies. It gives

$$\sum_a d_n(a)\,\overline{\chi}(a)=[1+\chi(n)-\chi(n+1)]\,B_{1,\overline{\chi}}.$$

Summing all $|G|=b$ slices: $\sum_{n\in G}\sum_a d_n(a)\,\overline{\chi}(a)=B_1[b-\overline{S_G(\chi)}]$.

Step 4 (combine). $\phi\,\hat{S}^\circ=0+(-bB_1)+B_1[b-\overline{S_G}]-0=-B_1\cdot\overline{S_G}$. □

Lemma 3. *For primitive χ modulo m and* $\gcd(n,m)=1$*:*

$$\sum_a \overline{\chi}(a)\,\{na/m\}=\chi(n)\,B_{1,\overline{\chi}}.$$

Proof. The substitution $a\mapsto n^{-1}a$ permutes the units and gives

$$\chi(n)\sum_a\overline{\chi}(a)\,\{a/m\}=\chi(n)\,B_{1,\overline{\chi}},$$

since $\{a/m\}=a/m$ for $1\le a\le m-1$. □

All identities involving F° and $P(s,\chi)=\sum_p\chi(p)/p^s$ are understood up to the finite contribution of primes $p\le m$, which is irrelevant for convergence.

4. The L-Encoding

The Bernoulli number is the L-function in disguise.

Corollary 4. $|\hat{S}^\circ(\chi)|=(b/\pi\phi)\,|L(1,\chi)|\cdot|S_G(\chi)|$.

Proof. By the generalized Bernoulli–L-value formula for odd primitive characters [1], $|B_{1,\overline{\chi}}| = (b/\pi)\,|L(1,\chi)|$. □

Corollary 5 (Short partial sum form)**.** *By the diagonal reduction (Lemma 1),*

$$|\hat{S}^\circ(\chi)| = \frac{2b}{\pi\phi}\,|L(1,\chi)| \cdot \Big|\sum_{k=1}^{b-1} \overline{\chi}(k)\Big|.$$

Remark 6 (Composite bases). At base 5, $|\sum_{k=1}^{4} \overline{\chi}(k)| = (\sqrt{5}/2)\,|B_{1,\overline{\chi}}|$ for all 8 primitive odd characters modulo 25, giving $|\hat{S}^\circ| \propto |L(1)|^2$. Computation in composite bases $b \in \{6, 10, 12\}$ shows $2\,|\sum_{k=1}^{b-1} \overline{\chi}(k)| = |B_{1,\overline{\chi}}|$ for all primitive odd χ, giving $|\hat{S}^\circ(\chi)| = |B_{1,\overline{\chi}}|^2/\phi(b^2)$ with a single proportionality constant $1/(\pi\,\varphi(b))$. A proof of this identity for composite bases is open.

Remark 7. For quadratic odd primitive characters χ_D with fundamental discriminant $D < -4$, $|L(1,\chi_D)| = \pi h(D)/\sqrt{|D|}$. When such characters appear among the primitive odd characters modulo m, the decomposition theorem specializes to class-number data.

5. Moment Bounds

Theorem 8 (Moment identity)**.** *For prime b and $m = b^2$:*

$$\sum_{\substack{\chi \bmod m \\ \chi \text{ prim. odd}}} |L(1,\chi)|^2 \cdot |S_G(\chi)|^2 = \frac{\pi^2\phi(m)}{b^2} \sum_{\substack{a \bmod m \\ \gcd(a,m)=1}} |S^\circ(a)|^2.$$

Proof. Parseval's identity on $(\mathbb{Z}/m\mathbb{Z})^\times$ gives

$$\sum_{\chi} |\hat{S}^\circ(\chi)|^2 = \frac{1}{\phi}\sum_{a} |S^\circ(a)|^2.$$

By the antisymmetry theorem [5], $\hat{S}^\circ(\chi) = 0$ for even χ. For imprimitive odd χ induced from a character ψ modulo b, both $S_G(\chi) = 0$ and $\hat{S}^\circ(\chi) = 0$.

For S_G: since $G = \{r(b{+}1) : 0 \le r \le b{-}1\}$ and χ depends only on residues modulo b, we have $\overline{\chi}(r(b{+}1){+}1) = \overline{\psi}(r{+}1)$ and $\overline{\chi}(r(b{+}1)) = \overline{\psi}(r)$, so $S_G(\chi) = \sum_{r=0}^{b-1}[\overline{\psi}(r{+}1) - \overline{\psi}(r)] = 0$ (telescoping, using $\psi(0) = \psi(b) = 0$).

For $\hat{S}^\circ$: on each coset $\{a : a \equiv k \pmod b\}$, the imprimitive $\overline{\chi}$ is constant, while S° has mean zero on that coset by definition of centering. Each coset contributes zero, hence $\hat{S}^\circ(\chi) = 0$. Only primitive odd characters survive. Substituting the L-encoding (Corollary 4) and rearranging gives the identity. □

The right side is computable: S° is explicitly determined on the $\phi(m)$ units, and the weights $|S_G(\chi)|^2$ on the left are determined by the collision geometry.

Remark 9. At base 5, the identity $|S_G| = (5\sqrt{5}/\pi)\,|L(1)|$ (verified for all 8 primitive odd characters) converts the moment identity into $\sum |L(1,\chi)|^4 = c_5 \sum |S^\circ(a)|^2$: a fourth moment of L-function special values, computable from the collision invariant.

Corollary 10 (Conditional cross-moment)**.** *For prime b and $m = b^2$: if every primitive odd L-function modulo m satisfies $L(s,\chi) \neq 0$ for $\mathrm{Re}(s) > \sigma_0$, then for real $s > \sigma_0$ the weighted moment*

$$\frac{1}{\phi(m)} \sum_{\chi \text{ prim. odd}} |B_{1,\overline{\chi}}| \cdot |S_G(\chi)| \cdot |P_{>m}(s,\chi)|$$

is bounded below by $|F^\circ_{>m}(s)|$, where $P_{>m}(s,\chi) = \sum_{p>m} \chi(p)/p^s$ and $F^\circ_{>m}(s) = \sum_{p>m} S^\circ(p)/p^s$.

Proof. The zero-free hypothesis ensures that $P_{>m}(s,\chi)$ is well-defined for each primitive odd χ. Using the transform expansion from [5] and the vanishing of even and imprimitive odd coefficients established above, $F^\circ_{>m}(s) = \sum_\chi \hat{S}^\circ(\chi)\, P_{>m}(s,\chi)$ with only primitive odd characters contributing. The triangle inequality gives the bound. □

Remark 11. The bound relates $|L(1,\chi)|$ (from B_1, at the edge of the critical strip) to $|P(s,\chi)|$ (in the strip). Since $P = \log L - H$ and H is small for large b, this is approximately a bound involving $|\log L(s,\chi)|$.

6. The Correlation Decay

The decomposition theorem shows what the collision spectrum encodes. What remains is how faithfully. The partial sum $P = \sum_{k=1}^{b-1} \overline{\chi}(k)$ has an exact decomposition via the classical identity for short character sums [2, 3]. After normalizing by the Gauss sum:

$$\widetilde{P} = L(1,\overline{\chi}) + \Delta(\chi),$$

where the packet

$$\Delta(\chi) = \frac{i}{\varphi(b)} \sum_{\substack{\xi \bmod b \\ \xi(-1)=1,\ \xi\neq 1}} \tau(\overline{\xi})\, L(1,\xi\overline{\chi})$$

is a sum of twisted L-values.

Computation using [6] suggests three properties of the packet. Its magnitude relative to $L(1)$ appears constant (mean $|\Delta|/|L| \approx 1.08$). Its phase relative to $L(1)$ appears uniformly distributed. And the

b	mean $\lvert\Delta\rvert/\lvert L\rvert$	std	std $\cdot \log b$
5	0.80	0.65	1.05
7	1.03	0.65	1.26
13	1.11	0.50	1.28
19	1.10	0.42	1.23
31	1.07	0.33	1.12
43	1.06	0.29	1.09

TABLE 1. The packet Δ appears to have constant magnitude relative to $L(1)$ (mean ≈ 1.08) and uniform phase ($\langle\cos\theta\rangle = 0.000$). The observed standard deviation of $|\Delta|/|L|$ is consistent with decay as $c/\log b$.

standard deviation of the ratio $|\Delta|/|L|$ is consistent with decay as $c/\log b$ (Table 1).

This observed variance decay is consistent with the measured correlation between $|\widetilde{P}|$ and $|L(1)|$: at small b, the high variance of $|\Delta|/|L|$ allows $|\widetilde{P}| = |L + \Delta|$ to track $|L|$; at large b, the low variance makes $|\Delta|/|L|$ nearly constant, and the uniform phase washes out the L-specific information.

7. Remarks

The decomposition theorem explains the observations of the companion paper [5]: the anti-correlation between collision coefficients and prime character sum magnitudes, the absence of the principal-character term, and the mod-3 structure are all consequences of the factor B_1 in the identity $\hat{S}^\circ = -B_1 \cdot \overline{S_G}/\phi$.

The proof uses three classical ingredients (the slice formula, the Bernoulli identity, and the vanishing of primitive character sums over cosets) applied to a new object (the collision invariant). The diagonal reduction (Lemma 1) eliminates S_G in favor of the short partial sum $\sum_{k=1}^{b-1} \overline{\chi}(k)$, a simpler and more classical object. The decomposition identity is exact for every primitive odd character at every base.

The analytic continuation from [5] combined with the decomposition gives $\mathcal{F}^\circ(s) = -(1/\phi) \sum B_1 \cdot \overline{S_G} \cdot [\log L(s,\chi) - H(s,\chi)]$. Near a zero ρ of $L(s,\chi_0)$, the coefficient of the divergent term involves $L(1,\chi_0)$ through B_1: the singularity at depth s is weighted by the health at $s = 1$.

What remains open is the proof of the variance decay $\mathrm{std}(|\Delta|/|L|) \sim c/\log b$ from the Apostol formula: this would establish the correlation

decay between partial character sums and L-function special values as a theorem.

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